

## CHIANG MAI INTL, Thailand

WMO: 483270

Lat: 18.7714N

Lon: 98.9692E

Elev: 1030

StdP: 14.16

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 55.6       | 58.1 | 42.7                          | 42.0 | 85.3 | 46.3 | 48.2 | 82.4 | 12.0                  | 76.7 | 10.0 | 78.7 | 0.7                   | 260  | 0.298 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 22.8                   | 100.5           | 73.0 | 98.5 | 72.8 | 96.4 | 72.6 | 80.4                | 89.5 | 79.5 | 88.4 | 78.8 | 87.7 | 6.2                  | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.6                            | 154.1 | 85.1 | 77.3 | 147.6 | 83.0 | 76.9 | 145.4 | 82.4 | 45.1          | 90.0 | 44.1 | 88.7 | 43.4 | 88.3 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 13.8          | 11.3 | 9.5 | DB                         | 51.1 | 103.0              | 4.2 | 2.4                                                | 48.1 | 104.7      | 45.6 | 106.1      | 43.2 | 107.5      | 40.2 | 109.2 | (4) |
| (5)               |      |     | WB                         | 49.1 | 82.4               | 3.8 | 1.7                                                | 46.4 | 83.6       | 44.1 | 84.6       | 42.0 | 85.6       | 39.2 | 86.9  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 80.2        | 73.1  | 76.8  | 81.8  | 85.9  | 84.2  | 83.3  | 82.2  | 81.6  | 81.7  | 80.7  | 77.5  | 73.4  | (6)  |  |
|                                                                        | DBStd       | 4.98        | 3.22  | 3.29  | 3.33  | 3.41  | 3.68  | 2.41  | 2.46  | 2.15  | 2.08  | 2.39  | 3.44  | 4.03  | (7)  |  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 3           | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | (9)  |  |
|                                                                        | CDD50       | 11014       | 715   | 750   | 985   | 1076  | 1061  | 1000  | 997   | 978   | 950   | 951   | 825   | 726   | (10) |  |
|                                                                        | CDD65       | 5545        | 252   | 330   | 521   | 626   | 596   | 550   | 532   | 513   | 500   | 486   | 375   | 264   | (11) |  |
|                                                                        | CDH74       | 56226       | 2077  | 3464  | 6227  | 8097  | 6796  | 5895  | 5290  | 4712  | 4510  | 4258  | 2996  | 1904  | (12) |  |
|                                                                        | CDH80       | 24615       | 720   | 1694  | 3462  | 4644  | 3335  | 2411  | 1903  | 1551  | 1592  | 1614  | 1109  | 580   | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.1         | 2.0   | 2.5   | 3.2   | 4.1   | 4.2   | 4.0   | 3.7   | 3.3   | 3.0   | 2.8   | 2.4   | 2.2   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 46.1        | 0.4   | 0.3   | 0.7   | 2.1   | 6.5   | 5.2   | 6.1   | 8.8   | 8.6   | 4.9   | 2.0   | 0.5   | (15) |  |
|                                                                        | PrecMax     | 62.1        | 3.1   | 2.6   | 7.1   | 8.4   | 16.2  | 13.2  | 17.6  | 16.7  | 20.9  | 9.8   | 13.1  | 3.0   | (16) |  |
|                                                                        | PrecMin     | 29.1        | 0.0   | 0.0   | 0.0   | 0.0   | 0.5   | 0.6   | 2.0   | 2.6   | 2.5   | 0.5   | 0.0   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 8.6         | 0.7   | 0.6   | 1.2   | 1.8   | 3.4   | 2.5   | 2.9   | 3.3   | 3.5   | 2.4   | 2.4   | 0.8   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 90.1  | 96.4  | 100.6 | 104.0 | 102.4 | 97.1  | 95.3  | 93.0  | 93.6  | 93.4  | 91.8  | 89.7  | (19) |  |
|                                                                        |             | MCWB        | 68.5  | 68.2  | 70.1  | 72.3  | 75.3  | 77.5  | 77.6  | 77.9  | 78.0  | 77.1  | 74.4  | 72.4  | (20) |  |
|                                                                        | 2%          | DB          | 88.1  | 93.8  | 98.6  | 101.9 | 98.6  | 94.6  | 92.6  | 91.1  | 91.6  | 91.4  | 89.9  | 87.6  | (21) |  |
|                                                                        |             | MCWB        | 67.9  | 67.4  | 69.6  | 72.9  | 76.1  | 77.5  | 77.2  | 77.6  | 77.4  | 76.2  | 73.8  | 71.0  | (22) |  |
|                                                                        | 5%          | DB          | 86.4  | 91.8  | 96.8  | 99.6  | 95.6  | 91.8  | 90.2  | 89.4  | 89.7  | 89.7  | 88.2  | 85.8  | (23) |  |
|                                                                        |             | MCWB        | 67.4  | 67.0  | 69.2  | 72.9  | 76.4  | 77.4  | 76.9  | 77.2  | 77.1  | 75.7  | 73.2  | 70.2  | (24) |  |
|                                                                        | 10%         | DB          | 84.4  | 89.8  | 94.8  | 97.0  | 93.2  | 89.9  | 88.2  | 87.6  | 87.9  | 88.0  | 86.3  | 83.8  | (25) |  |
|                                                                        |             | MCWB        | 67.0  | 66.8  | 69.0  | 73.3  | 76.4  | 77.2  | 76.6  | 77.0  | 76.8  | 75.5  | 72.5  | 69.2  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 73.4  | 74.1  | 76.3  | 79.8  | 81.4  | 82.5  | 80.9  | 80.8  | 80.9  | 79.6  | 78.2  | 75.9  | (27) |  |
|                                                                        |             | MCDB        | 83.4  | 88.5  | 91.4  | 94.5  | 93.2  | 90.2  | 89.4  | 87.8  | 88.6  | 88.5  | 86.9  | 85.1  | (28) |  |
|                                                                        | 2%          | WB          | 71.8  | 72.2  | 74.6  | 78.0  | 79.6  | 80.1  | 79.2  | 79.5  | 79.4  | 78.4  | 76.5  | 73.9  | (29) |  |
|                                                                        |             | MCDB        | 81.3  | 85.9  | 90.0  | 93.1  | 91.4  | 88.9  | 87.3  | 86.7  | 87.1  | 87.2  | 85.4  | 82.9  | (30) |  |
|                                                                        | 5%          | WB          | 70.5  | 70.8  | 73.4  | 76.8  | 78.6  | 79.0  | 78.3  | 78.6  | 78.6  | 77.6  | 75.4  | 72.5  | (31) |  |
|                                                                        |             | MCDB        | 80.7  | 84.4  | 89.0  | 91.8  | 89.8  | 87.8  | 85.9  | 85.7  | 86.0  | 85.9  | 83.9  | 81.0  | (32) |  |
|                                                                        | 10%         | WB          | 69.2  | 69.4  | 72.1  | 75.6  | 77.7  | 78.1  | 77.7  | 77.9  | 77.8  | 76.7  | 74.4  | 71.1  | (33) |  |
|                                                                        |             | MCDB        | 79.8  | 83.4  | 86.7  | 89.8  | 88.1  | 86.2  | 84.8  | 84.3  | 84.7  | 84.5  | 82.3  | 79.2  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 24.8  | 28.0  | 26.4  | 22.8  | 17.3  | 14.6  | 13.0  | 12.8  | 14.1  | 16.0  | 18.8  | 21.2  | (35) |  |
|                                                                        |             | MCDBR       | 27.2  | 30.3  | 29.2  | 25.8  | 21.1  | 17.3  | 16.0  | 15.4  | 16.4  | 18.2  | 20.2  | 22.5  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 10.2  | 10.3  | 9.9   | 8.4   | 7.5   | 6.3   | 5.7   | 5.9   | 6.2   | 6.4   | 7.3   | 8.6   | (37) |  |
|                                                                        |             | MCWBR       | 22.4  | 25.1  | 24.1  | 22.2  | 18.1  | 15.5  | 13.9  | 13.7  | 14.8  | 16.3  | 17.6  | 19.2  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.434 | 0.510 | 0.679 | 0.677 | 0.507 | 0.442 | 0.429 | 0.445 | 0.476 | 0.515 | 0.441 | 0.436 | (40) |  |
|                                                                        |             | taud        | 2.206 | 2.001 | 1.619 | 1.687 | 2.162 | 2.380 | 2.413 | 2.351 | 2.243 | 2.092 | 2.262 | 2.227 | (41) |  |
|                                                                        | Ebn at Noon |             | 266   | 252   | 215   | 217   | 255   | 270   | 273   | 271   | 263   | 247   | 263   | 262   | (42) |  |
|                                                                        |             | Edh at Noon | 43    | 56    | 84    | 79    | 49    | 39    | 38    | 40    | 44    | 50    | 40    | 41    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1465  | 1715  | 1794  | 1904  | 1698  | 1472  | 1366  | 1353  | 1434  | 1408  | 1454  | 1369  | (44) |  |
|                                                                        | RadStd      |             | 74    | 61    | 114   | 111   | 144   | 102   | 104   | 84    | 61    | 78    | 104   | 83    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.19 | +1.78   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +488  | +491  |
| (47) Regional (15 neighbors) |  | +0.50 | N/A     | N/A    | N/A     | N/A   | +0.27 | N/A         | N/A   | +248  | +250  |

Nomenclature: See separate page